Staff Messaging (1-1Chat)

0) Scope & Assumptions

- **Users:** Org Admins, School Admins, Teachers/ Staff
- **Purpose:** Enable secure, role-based communication between Admins & staff
- **Tenancy:** Messaging is scoped to the selected organization and school context.
- **Session:** Browser-based SPA; JWT or session cookie.
- **Timezone:** Use school's timezone for timestamps.

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1) User Journey

Persona: Teacher/Admin
Actions:

- Navigate to **Communication → Staff Messaging**
- View inbox with conversation threads.
- Compose a new message (bulk) OR send message in a single chat.
- Track delivery and read status.

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2) Core Functionalities

2.1 Inbox & Conversation View

- Sidebar lists conversations:
  - Staff name, Photo/ initials + last message preview with timestamp
  - Unread badge count.
- Search Staff (Active & upcoming Staff): Should persist until changed
- (edits /deletes will cascade & update in db)

Field Name	Data Type	Default Value	Validations	Additional Notes
Search Staff		chronological order.		
Classroom Filter	Dropdown Multiselect	Empty (showing all	Must match active classrooms	Multi-select; displayed in filter panel

		Active staff in all classes)		
Designation Filter		Empty (showing all Active Staff)	Must match predefined Designations	Multi-select; displayed in filter panel
Status Filter	Enum	Active	Allowed values: Active, Upcoming	Default: Active; Allow upcoming & Inactive students only if filter applied
clear all	Button			Clears all filter (no need to click apply again)

- Toggle: “Show only unread.” (filters only conversations with at least one unread message)
- Clicking a conversation opens threaded view:
  - Messages in chronological order.
  - Each message shows sender name, timestamp.
  - **Compose & send message :**Inside a conversation thread, user can compose and send a message **specific to that chat** (single recipient context).

Field Name	**Data Type**	**Default Value**	**Validations**	**Additional Notes**
Message Text	String	Empty	Required; Max 2000 characters	Supports **Rich Text** (bold, italic, underline, line spacing when enter etc.) , Emojis & URLs
Character Counter	Integer	6000	Decrements as user types	Displayed below text box
Attachement	file	None	optional, JPEG, PNG, HEIC, PDF & Docs files allowed each	Photo visible directly in chat. , PDF or doc is like file (sameas in

			file<10mb, Multiple allowed	whatsapp) multiple photo or file view same as whatsapp
Send Button	Boolean	Disabled	Enabled only if message text is not empty	Primary CTA

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2.2 Compose New Message (Bulk Action)

a. **Flow:** Click **New Message**.

b. **Step 1:** Select recipients:
   - Search bar for staff names.
   - Filters: Classroom, Designation , Status (Active/Inactive).
   - **Default:** Show only active staff; exclude inactive; allow upcoming staff.
   - “Select All” option.
   - Ability to remove selected recipients.

Field Name	**Data Type**	**Default Value**	**Validations**	**Additional Notes**
Search Student	String	Empty	Max length: 100 chars	Supports partial match; case-insensitive search
Classroom Filter	Dropdown Multiselect	Empty (showing all students in all classes)	Must match active classrooms	Multi-select; displayed in filter panel
Designation Filter		Empty (showing all students)	Must match predefined Designation	Multi-select; displayed in filter panel
Status Filter	Enum	Active	Allowed values: Active, Upcoming	Default: Active; Inactive students excluded

Staff List	Array, link object with photo /initials	Active Staff only	At least one staff must be selectable to go to next step	Upcoming staff can appear if filter selected for that; inactive hidden
Clear all				clears all the filters to defaults
Select All Checkbox	Boolean	False	N/A	Selects all currently filtered students
Selected Recipients	Array	Empty	At least one recipient required before sending	Ability to remove recipients individually

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Step 2 – Message Composition

- Rich text editor (bold, italic, underline) with emojis and optional attachment.
- URL-compatible text (clickable links).
- Character counter (max 6000 chars).
- **Send** button enabled only if:
  - At least one recipient selected.
  - Message text is not empty.

Field Name	**Data Type**	**Default Value**	**Validations**	**Additional Notes**
Message Text	String	Empty	Required; Max 2000 characters	Supports **Rich Text** (bold, italic, underline, line spacing when enter

				etc.) , Emojis & URLs
Character Counter	Integer	6000	Decrements as user types	Displayed below text box
Attachement	file	None	optional, JPEG, PNG, HEIC, PDF & Docs files allowed <10mb	Photo visible directly in chat. , PDF or doc is like file (sameas in whatsapp)
Go Back	Boolean	Active		Goes back to student selection
Send Button	Boolean	Disabled	Enabled only if message text is not empty AND ≥1 recipient selected	Primary CTA

- **Rich Text & URL support** in message composer.
- **Default filter behavior:** Active students only; exclude inactive; allow upcoming.
- **Validation:** Send button logic tied to both steps.

2.3 Message Actions

- On sent messages:
  - **Copy** text.
  - **Edit** message.
  - **Delete** message.
  - **Info / Status** → Opens “Message Status” popup:
    - Original message content.
    - Delivered To: list of recipients.
    - Read By: list of recipients who read the message.
- Read receipts visible per message.

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3) Filter Functionality

- Filter icon opens panel with:
  - Classroom filter (multi-select).

    - Designation filter (multi-select).
    - Status filter (Active, Upcoming).
    - Search filter for staff names.
- Default: Active staff only.

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4) Field Specifications

Field Name	Data Type	Default Value	Validations	Notes
Message Text	String	Empty	Max 2000 chars	Rich text + URL support
Recipients	Array	Empty	At least 1	Selected via filters/search
Filters	Enum	Active only	Classroom/Tag/Status	Optional

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5) Acceptance Criteria

- **AC1:** Inbox shows all conversations sorted by latest message.
- **AC2:** Unread badge count updates in real-time.
- **AC3:** “Send” disabled until message text and recipient(s) are provided. & once pressed disabled again until action is completed
- **AC4:** Delivery status accurate within 2 seconds of sending.
- **AC5:** Read receipts update within 5 seconds after recipient views message.
- **AC6:** Filters return correct staff list; default excludes inactive.
- **AC7:** Character counter decrements as user types.
- **AC8:** Rich text formatting and clickable URLs supported in composer.
- **AC9:** Single-chat composer sends message only to that conversation’s recipient.
- **AC10:** Delete action shows deleted message in thread (soft delete with text " This message has been deleted").
- AC14: Adding/ editing /removing classroom of the staff should update in chats accordingly
- AC16: All Cross platform actions (web & mobile app) Should perform exactly the same.
- AC17: Alerts & notifications should be as per alert sheet `https://docs.google.com/spreadsheets/d/1_oBkliWUoZ_aX-OeDs0f_HfbRcym9fAEephuqanm9WM/edit?pli=1&gid=0#gid=0` & Private (https://app.clickup.com/9017410820/docs/8cqnh84-10057/8cqnh84-9217)
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6) Security Requirements

- Role-based access: Only authorized staff can send messages.
- Validate recipient IDs server-side.
- Audit logs: userId, messageId, timestamp, orgId, siteId, action (send/edit/delete).

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7) Accessibility & Localization

- Keyboard navigation for all fields.
- ARIA labels for message status popup.
- WCAG 2.1 AA compliance.
- All strings i18n-ready.

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8) Non-Functional Requirements

- **Performance:** Inbox load < 500 ms; message send < 800 ms.
- **Scalability:** Support up to 10K schools 2M users without UI lag.
- **Reliability:** 99.9% uptime for messaging service.

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9) QA Test Matrix

- **Happy Path:** Compose → Send → Delivery → message actions (copy, edit,Read status, delete).
- **Edge Cases:** Empty message, no recipients, API failure.
- **Filters:** Classroom/designation/status combinations.
- **Unread toggle:** Shows only unread threads.
- **Rich text:** Bold, italic, links render correctly.

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10) Gherkin Scenarios

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Feature: Messaging
Scenario: Send message in single chat
  Given I am viewing a conversation thread
  When I type ”Hello” in the composer
  And click Send
  Then the message appears in that thread only

Scenario: Bulk message excludes inactive staff
  Given I click New Message
  When I view staff selection
  Then only active staff are shown
```